ASSIGNMENT 4

Q4: Custom Form Builder

Instructions: Save JavaScript-only solutions as `.js` files and run with Node.js. For form questions, save the full code as `.html` and open it in a browser. No frameworks are used.

Solution

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>Custom Form Builder</title><style>body{font-family:Arial;max-width:450px;margin:30px auto}label{display:block;margin:12px 0}in
<body><h1>Custom Form Builder</h1><div id="formContainer"></div><h2>Submitted Data</h2><pre id="output">Fill and submit the form.</pr
<script>
class FormBuilder {
 constructor(fields,containerId){this.fields=fields;this.container=document.getElementById(containerId);}
 createForm(){
  const markup=this.fields.map((field,i)=>{
   const type=["text","email","number","password"].includes(field.type)?field.type:"text";
   const safeLabel=field.label.replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;","'":"&#39;"}[c]));
   return `<label for="field-${i}">${safeLabel}<input id="field-${i}" type="${type}"></label>`;
  }).join("");
  this.container.innerHTML=`<form id="dynamicForm">${markup}<button>Submit</button></form>`;
  document.getElementById("dynamicForm").addEventListener("submit",e=>{
   e.preventDefault();document.getElementById("output").textContent=JSON.stringify(this.getFormData(),null,2);
  });
 }
 getFormData(){
  const data={};this.fields.forEach((field,i)=>data[field.label]=document.getElementById(`field-${i}`).value);
  return data;
 }
}
new FormBuilder([{type:"text",label:"Username"},{type:"email",label:"Email"},{type:"password",label:"Password"}],"formContainer").cre
</script></body></html>
```

Expected result / testing

Open in a browser, enter values and submit. The form data is displayed as a JSON object.